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Polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) architecture for optical communication links

Abstract

A polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) architecture for optical communication links has a receiver section that polarization un-rotates a signal from a fiber optic cable into first and second polarized optical signals; recovers a polarization of the first and second optical signals based on a received polarization recovery signal that is based on a pilot signal measurement signal; demodulates the first optical signal into optical QPSK data and pilot tone signals; demodulates the second optical signal into an optical modulating laser light; splits the first and second optical signals into optical QPSK quadrature signals; converts the optical QPSK quadrature signals into electrical QPSK quadrature signals; detects a polarization of the pilot tone signal and outputs the pilot signal measurement signal polarization recovery signal based on the detected polarization.

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Background/Summary

RELATED APPLICATION INFORMATION (1) This patent is a continuation application from co-pending U.S. patent application Ser. No. 18/307,717, filed Apr. 26, 2023, which claims priority from U.S. Provisional Patent Application No. 63/334,877, titled, POLARIZATION-MULTIPLEXED SELF-HOMODYNE ANALOG COHERENT (PM-SH-ACD) ARCHITECTURE FOR OPTICAL COMMUNICATION LINKS, filed Apr. 26, 2022, all of which are incorporated herein by reference.

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BACKGROUND

Field

(2) This disclosure relates to a polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) architecture for optical communication links.

Description of the Related Art

(3) Optical communication is an important part of modern communication techniques due to the excessive bandwidth of the light spectrum. Theoretically, optical communication has much higher system throughput than its radio frequency (RF) communication counterpart. Therefore, it finds many applications and facilitates our lives. Some typical optical communication scenarios include optical fiber communication, free-space optical communication, and visible light communication.

(4) Some architectures for optical communication links employ coherent modulation/demodulation, whereby both phase and amplitude of light are modulated to transmit data across an optical fiber channel. In coherent communication it is necessary to perform carrier recovery (CR) to demodulate the signal at the receiver. CR consists of phase-locking an unmodulated carrier to the received modulated signal.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a block diagram of a system for polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) optical communications having polarization recovery (PR) and carrier recovery (CR) prior to wavelength based demultiplexing.
- (2) FIG. 1B is a block diagram of a system for PM-SH-ACD optical communications having PR prior to and CR after wavelength based demultiplexing.
- (3) FIG. 1C is a block diagram of a system for PM-SH-ACD optical communications having PR and CR after wavelength based demultiplexing.
- (4) FIG. 2 is a schematic diagram **200** of combined PR and CR block that includes 8 phase shifters and 4 phase couplers.
- (5) FIG. 3 is a schematic diagram of an architecture for PM-SH-ACD optical communications having PR and CR prior to wavelength based demultiplexing.
- (6) FIG. 4 is a schematic diagram of a transmitter system having a laser and coherent Mach-Zehnder modulators (MZMs) that are backwards compatible with intensity modulated direct detect (IMDD) formats.
- (7) FIGS. 5A-5C are schematic diagrams of receiver systems **501 502** and **503** for PM-SH-ACD optical communications that are backwards compatible with IMDD formats.
- (8) FIGS. 6A-6C are block diagrams of systems for PM-SH-ACD optical communications that are backwards compatible with IMDD formats.
- (9) FIG. 7 shows a flow chart of a process flow for PM-SH-ACD optical communications.
- (10) FIG. 8A is an example of equations for how to obtain error signal $e(t)$ for polarization closed-loop control block diagram and of components of a receiver.
- (11) FIG. 8B is an example of Jones matrices for proportional, integral, derivative (PID) tuning of the phase shifters and couplers of the PR and CR block.
- (12) Throughout this description, elements appearing in figures are assigned three-digit or four-digit reference designators, where the two least significant digits are specific to the element and the one or two most significant digit may be the figure number where the element is first introduced or fabricated. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described or subsequently-described element having the same reference designator or two least significant digits.

DETAILED DESCRIPTION

- (13) Described herein is an implementation of an architecture for an optical communication link that employs coherent modulation/demodulation, whereby both phase and amplitude of light are modulated to transmit data across an optical fiber channel. In this coherent communication it is necessary to perform carrier recovery (CR) to demodulate the signal at the receiver. The CR may consist of phase-locking an unmodulated carrier to the received modulated signal.
- (14) One approach to carrier recovery in an optical coherent link having optical fiber channels is based on transmitting an unmodulated carrier in addition to the modulated optical signal along a parallel channel. One example of this consists of transmitting the unmodulated carrier in an orthogonal polarization on the same fiber as the modulated signal. This can be referred to as polarization-multiplexed self-homodyne (PM-SH) detection. In PM-SH, in addition to CR, the original transmitted state-of-polarization (SOP) needs to be recovered. The recovery of the initial transmitted SOP may be referred to as polarization recovery (PR). In coherent fiber optic links, CR and PR may be done using digital signal processing (DSP). Alternatively, both CR and PR may be separately demonstrated in the analog circuit domain. In analog CR, a circuit such as a Costas Loop may be employed, with a phase-frequency detector (PFD) and a loop filter that feeds back to a local oscillator to phase lock it to the carrier of the received modulated signal. In analog PR, a marker or pilot tone can be used to recover the original SOP at the TX.
- (15) The following describes architectures, systems, transmitters, receivers and processes for

polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) optical communications. These PM-SH-ACD optical communications have three key differentiating innovations and/or advantages.

(16) First, both PR and CR may be performed in the analog domain in a wavelength-multiplexed link without the need for dedicated high-frequency circuits, such as those running at 10 GHz RF. Both PR and CR may be done by maximizing the power of a pilot tone introduced to one or both of the quadratures (e.g., sine and cosine wave signals) of one of the two transmitted polarizations (e.g., 0 degree optical signal polarization and 90 degree optical signal polarization).

(17) Two, the use of this analog PR and CR approach may enable the use of conventional commercially-available intensity-modulation direct-detection (IMDD) analog circuits—drivers and transimpedance amplifiers (TIAs)—and digital signal processors (DSPs). For example, for an 800 Gb/s 4-wavelength 50 Gbaud 16QAM implementation, the analog circuits and DSP that are employed can be the same ones that are developed for 400 Gb/s 4-wavelength (or 4-fiber) 100 Gbaud pulse amplitude modulation (PAM4) IMDD links. The use of commercially available analog circuits and DSPs from prior generation IMDD links can significantly reduce latency, power dissipation, and cost of optical transceivers.

(18) Three, the use of this analog PR and CR approach offers backwards compatibility. Some single-fiber architectures or transceivers may transmit a total data rate of 400 Gb/s PAM4 IMDD. Specifically, a 400G FR4 standard may transmit 400 Gb/s using 4 wavelengths. An 800 Gb/s architecture herein may be backwards compatible with the 400G FR4. Some DSP-based coherent transceivers may not be backwards compatible. This is because the PR is done in the digital domain, and thus the waveguide polarization dependent losses result in a penalty which may render IMDD operation unfeasible. Furthermore, 1.6 Tb/s and 3.2 Tb/s solutions described herein may be backwards compatible with the proposed 800 Gb/s and 1.6 Tb/s solutions, respectively.

(19) Description of Apparatus

(20) FIG. 1A is a block diagram of a system **100** (e.g., an architecture) for polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) optical communications having polarization recovery (PR) and carrier recovery (CR) at block **134** prior to wavelength based demultiplexing at **136** and **137**. System **100** includes a transmitter section or transmitter (TX) **110** transmitting optical signal **182** to fiber optic cable **180** from which a receiver section or receiver (RX) **130** receives signal **182**. A system, an architecture or components thereof being described as “for” performing an action may also be “configured to” perform said action.

(21) In general, transmitter **110** converts input electrical data signals **117** from the DSP **116** into optical signals **125**. In the transmitter **110** there are one or more transmitter slices **120**, each corresponding to an optical wavelength of a different wavelength laser **112** (e.g., visual color of light) of each slice, which are multiplexed at MUX1 **126** and MUX2 **127**, after which polarization beam combiner (PBC) **128** forms polarization orthogonal optical signal **182** from the multiplexed signals. The optical signal **182** propagates across some length of optical fiber **180**. At the other end of the optical fiber **180**, a receiver (RX) **130** converts the optical signal **182** back into output electrical data signals **147** which include the same data as that of signals **117**. The receiver **130** also has one or more slices **140** (see also slices **162** and **172** of FIGS. 1B and 1C), each corresponding to a demultiplexed optical wavelength (e.g., of a different wavelength laser **112**).

(22) Transmitter **110** is an optical transmitting component including a transmitter digital signal processor (DSP) **116** sending electrical signals **117** to each of a number of transmitter slices **120**. Each TX slice **120** includes a laser **112**, a splitter **114**, drivers **118**, modulator **122** and SOA **124**. A first multiplexer (MUX1) is for receiving modulated optical signals **123** or **125** from all of the slices and sending a combination of those signals as part of signals **129** to a PBC **128**. A second multiplexer (MUX2) is for receiving unmodulated optical signals **115** from all of the slices and sending a combination of those signals as part of signals **129** to the PBC **128**. The PBC is for combining signals **129** into orthogonally polarized signals and sending them as signal **182** on

optical fiber **180**, such as a fiber optic cable.

(23) In system **100**, in transmitter **110**, there is a pilot tone (PT) signal **192** added to only one quadrature of one of the slices **120** by modulator **122**. Here, signal **192** is modulated onto one of quadrature I (cosine wave signals) or Q (sine wave signals) of the signals **121** that will be part of signals **123**, **125** and **129** of only the modulated one of the two transmitted polarizations (e.g., 0 degree optical signal polarization or 90 degree optical signal polarization) of signal **182**. In other cases, signal **192** can add to both of the quadratures of only the modulated one of the two transmitted polarizations.

(24) In system **100**, at the receiver **130**, the received signal **182** first goes to the polarization splitter rotator (PSR) **132**, then the PR and CR block **134**, and is then demultiplexed at DEMUX1 **136** and DEMUX2 **137**. In system **100**, the amplitude of the pilot tone signal **192** can be monitored at receiver **130** using: a tap **151** directed to a low-frequency photodiode (LF PD) **150** and a resistor (the resistor maybe in blocks **150** or in controller **158**); measuring a voltage signal of a dedicated transimpedance amplifier (TIA) **154**; and/or by measuring the voltage signal of one of the same TIAs **146** used for data transmission. All three options may provide an output voltage signal with an amplitude based on the amplitude of signal **192**. These voltage signal output options are depicted as signal **152** output from blocks **150**, signal **156** output from TIA **154** and signal **153** output from one of TIAs **146**, any one or more of which are represented as feedback signal **155**, which may be a pilot signal measurement signal **155**. Based on the signal **155**, the amplitude of the pilot tone signal **192** is measured using a microcontroller (μ C) **158**, which may perform a PR and CR algorithm for or of block **134**. This measurement and/or feedback signal **155** may be based on selecting any one or more of signals **152**, **156** and/or **153**. In some cases, the measurement is an aggregate or other combination of two or more of signals **152**, **156** and/or **153**.

(25) In more detail, each TX slice **120** contains a laser **112** that is operated at a constant output power at a particular wavelength (e.g., color) that is different for each Tx slice **120**. The constant laser output **113** is also referred to as continuous-wave (CW). Laser **112** may be a coherent light source for outputting a laser light **113** having one frequency and one phase to a splitter **114**.

(26) The laser output signal **113** is optically split into two paths by the splitter **114**. The power split signals **115** can be equal or unequal power among the two output paths. In some cases, splitter **114** is for creating two equal power outputs **115** of the laser light **113**. In FIG. 1A, the splitter's upper output signal is directed to an electro-optic modulator **122** and lower output signal is directed to an optical wavelength multiplexer (MUX2) **127**.

(27) An input data signal (not shown), which is composed of one or multiple parallel electrical data streams, is processed by a transmitter digital signal processor (DSP) **116**. The DSP output **117** signal can be or include one or several parallel electrical channels, such as the 4 parallel channels shown. There may be fewer or more than 4 parallel channels. Each DSP output channel is in one of various modulation formats, such as non-return to zero (NRZ) or 4-level pulse amplitude modulation (PAM4). The DSP outputs **117** are connected to the inputs of modulator drivers **118**, which may be voltage amplifiers **119** that amplify or QPSK modulate signal **117** into signals **121**. The modulator driver output electrical signals **121** are directed to electro-optic modulators **122**.

(28) The electro-optic modulators **122**, such as Mach-Zehnder modulators (MZM), convert the voltage signals **121** into an optical signal **123**. The modulators are biased such that the output optical signal **123** is phase-modulated in a format such as quadrature phase shift keying (QPSK) or 16-ary quadrature amplitude modulation (16QAM). In the case of a MZM, the modulator is biased at the null-transmission point to produce a phase-modulated signal **123**. In the case of a phase-modulated signal, the modulation may give QPSK phase values for two optical signals per quadrature, I+/- and Q+/- as optical signal **123**.

(29) Additionally, a marker or pilot tone (PT) signal **192** from tone generator **190** is added to one or two of the quadratures of modulators **122** for PR and CR at the receiver **130**, **160** or **170**.

Modulators **122** may add the PT signal **192** to only one quadrature, such as I- cosine wave signals

or Q- sine wave signals of the signals **121** of each of the slices **120**. In other cases, a signal **192** can add to both of the quadratures (e.g., sine and cosine wave signals) of the modulated one of the two transmitted polarizations. Each of the two tone signals **192** may be at different frequencies in order to distinguish them at receiver **130**. Modulator **122** may be an optical modulator for outputting an optical modulated data signal **123** (such as having 2 optical quadrature component signals (I+, Q+) of the output of the laser light **115** modulated with a data signal **121** and one or two pilot tone signal(s) **192** identifying the polarization of the modulated signal **123**. In some cases, data signal **121** may be QPSK or IMDD.

(30) While the data signal **117** or **121** may be in a 10 GHz range (e.g., 5-30 GHz, in the case of a 50 Gbaud embodiment), the PT signal **192** may be in the 1 MHz range (e.g., 0.5-10 MHz). The data signal **117** or **121** may be at a frequency that is at least 500× or 1000× the frequency of the signal **192**.

(31) The PT signal **192** may have an inconsequential amount of power as compared to the data signal **121** or **123**, such as with respect to the function of block **134**, DEMUXs **136** and **137**, hybrid **142**, PDs **144**, TIAs **146** and DSP **148**. The PT signal **192** may have between 1 and 10 percent of the power of signal **121** or **123**. It may have 5 percent of the power of signal **121** or **123**.

(32) The modulated signal **123** may then be amplified using an optical amplifier such as a semiconductor optical amplifier (SOA) **124** and is then directed as amplified signal **125** to an optical wavelength multiplexer (MUX1) **126**. SOA **124** may represent at least one SOA of transmitter **110** at and amplifying at least one of the signal **123** (as shown), the signal **115**, the signal **129** and/or the signal **119**. In some cases, SOA **124** does not exist.

(33) MUX1 **126** and MUX2 **127** are optical multiplexers, such as arrayed waveguide gratings (AWG). Multiplexer **126** takes multiple optical input signals **125**, one from each of the slices **120**, and combines the signals **125** into a single optical waveguide signal **129**. Multiplexer **127** takes multiple optical input signals **115**, one from each of the slices **120**, and combines the signals **115** into a single optical waveguide signal **119**. Each input signal **125** and **115** corresponds to one TX slice **120**, each of which utilizes a unique carrier wavelength of signal **115**. MUX1 **126** multiplexes the modulator output signals **125** of all the slices **120** and MUX2 **127** multiplexes the unmodulated carrier signals that correspond to the splitter's output signals **115** of all the slices **120**. MUX1 **126** may be a multiplexer for combining the optical modulated data signals **125** of the slices **120** into a combined optical modulated data signal **129**; and MUX2 **127** may be a multiplexer for combining the laser light signals **115** of the slices **120** into a combined optical carrier signal **119**.

(34) The optical output signals **129** and **119** of MUX1 **126** and MUX2 **127** are typically in the same polarization. In that case, polarization beam combiner (PBC) **128** rotates one of the multiplexer output signals **129** or **119** to an orthogonal polarization and then combines the signals **129** and **119** (one having orthogonal polarization as compared to the other). Rotating may be rotating one of signals **129** or **119** to have an optical polarization that is 90 degrees different than that of the other signal of signals **129** or **119**. The optical signal **182** at the output of the PBC **128** is thus a wavelength-multiplexed dual-polarization signal. This output signal **128** is then coupled to a fiber **180** for transmission to the receiver **130**. PBC **128** may be for polarization rotating by 90 degrees (e.g., orthogonally) one of the combined optical modulated data signal **129** or the combined optical carrier signal **119**; for combining into a polarized signal **182**, the polarization rotated one of and the other of the combined signals **129** and **119**; and for transmitting to a fiber optic cable **180** the polarized signal **182**.

(35) The modulated signal **125**, having PT signal **192** is in only one of the two transmitted polarizations of signal **182**. It may be in only signal **129** of signal **182** having a 0 degree optical polarization or a 90 degree optical polarization, while the other signal **119** has the other polarization. It may be in signal **129** while signal **119** has an orthogonal polarization compared to signal **129**.

(36) Cable **180** may be fiber optic or optical cables. It may be between a few feet and thousands of

miles in length. It may extend between ten miles and hundreds of miles.

(37) Receiver **130** is an optical receiving component including a receiver polarization splitter rotator (PSR) **132** for receiving from cable **180** and combining the orthogonal polarized signal **182** into parallel orthogonal signals **133** and **135** which have their polarity and carrier recovered at PR and CR (e.g., PR plus CR) block **134**, which outputs optically demodulated data signal and carrier signals **135**. PR and CR at block **134** is controlled or performed by microcontroller **158** which receives feedback signal **155** based on the amplitude and signs of the pilot tones signal **192**. Signals **135** are demultiplexed at DEMUX1 **136** and DEMUX2 **137** which send optical signals **139** to each of a number of receiver slices **140**. Receiver digital signal processor (DSP) **148** receives electrical signals **147** from each of the slices **140**. Each slice **140** may be an optical signal to electrical signal converter. In some cases, it may convert to QPSK or to IMDD.

(38) Each RX slice **140** includes a 90 degree optical hybrid **142**, photodiodes **144**, and TIAs **146**. Each RX slice **140** also includes a tap **151** directed to an LF PD **150** and a resistor in blocks **150** or controller **158**; and/or a dedicated TIA **154**. It also includes feedback signals **152**, **156** and/or **153**, any one or more of which are represented as feedback signal **155** to microcontroller (μ C) **158**.

(39) In more detail, the received optical signal **182** from cable **180** is coupled to the receiver **130** and directed to PSR **132**. The PSR **132** receives signal **182** with two orthogonal polarizations, which include all or parts of signals **129** and **119**, one orthogonally polarization to the other. The PSR **132** splits then rotates the two orthogonal polarizations to two same polarization signals **133** and **135** which the receiver waveguides are designed for. Rotating may be rotating one of polarization orthogonal signals **133** or **135** to have an optical signal polarization that is the same as that of the other signal of signals **133** and **135**.

(40) Signal **182** includes signals **129** and **119**, one orthogonally polarization to the other. Thus, each of signals **133** and **135** includes a part of both signals **129** and **119**, because the PSR **132** does not know the exact polarization of either or both of signals **129** and **119**, but has split off a randomly polarized pair of orthogonally polarization signals from signal **182**, and rotated one of those split off signals. Each of the randomly polarized pair of signals **133** and **135** will have part of signal **129** and part of signal **119**. The two output signals **133** and **135** are routed towards PR and CR block **134**. PSR **132** may be for receiving the polarized signal **182**; polarization un-rotating by 90 degrees the polarized signal **182** into same polarized optical signals **133** and **135**, each having some of the combined laser light signals **119** and some of the combined optical modulated data signals **129**.

(41) Polarization recovery of signals **129** and **119** from signals **133** and **135** happens at PR and CR block **134**. The state-of-polarization (SOP) of signal **182** undergoes random time-varying rotations as it propagates along fiber **180**. For that reason, it is necessary to de-rotate the received SOP to recover the transmitted signals **129** and **119**. The PR and CR block **134** receives the rotated SOP of signals **133** and **135**, and outputs the original transmitted SOP with phase-locked unmodulated carrier signal **119** and phase-locked modulated (with data signals **121**) carrier signal **129**. The PR and CR block **134** is or includes 8 phase shifters and 4 couplers; and is described further at FIG. 2. Block **134** may be a reset-free polarization controller and carrier recovery based on phase shifters.

(42) PR and CR block **134** includes a polarization recover (PR) component for recovering the polarization of the polarized optical signals **133** and **135** using a received polarization recovery signal **159** from the microcontroller **158**, to recover a polarization fixed optical signal **129** having the data signal **121** and pilot tone signal **192** optically modulated with the laser light **115** and to recover a polarization fixed optical signal having the laser light **115**, where signals **129** and **115** have the same fixed polarization.

(43) By implementing the closed-loop control described above, both PR and CR can be performed in the same block **134**, such as at the same time. PR and CR block **134** also includes a carrier recovery (CR) component for: (a) recovering the frequency and phase of the second output of the laser light from the polarization fixed second optical signal; (b) demodulating the polarization fixed first optical signal into a polarization and frequency fixed first optical signal having the data signal

and the pilot tone signal; and (c) demodulating the polarization fixed second optical signal into a polar, frequency and phase fixed second optical signal having or that is the second output of the laser light. Recovering the frequency and phase may be done using pairs of phase shifters and phase couplers as noted in FIG. 2; and/or be using the received carrier recovery signal (e.g., signal **159**) from the microcontroller **158**. In some cases, demodulating at (a) above is not performed. In some cases, the carrier recovery signal is or is based on a received polarization recovery signal.

(44) DEMUX1 **136** and DEMUX2 **137** are optical demultiplexers, such as arrayed waveguide gratings (AWG). The demultiplexers take a single optical input and separate it into multiple optical outputs, each corresponding to a different carrier wavelength. Each output corresponds to one RX slice of slices **140**. Demultiplexer **136** takes a single optical waveguide signal having combined data signals **129** for all slices **140** and splits signal **129** into multiple optical signals **125**, one for each of the slices **140**. Demultiplexer **137** takes a single optical waveguide signal having combined carrier signals **119** for all slices **140** and splits signal **119** into multiple optical signals **115**, one for each of the slices **140**. Each signal **125** and **115** corresponds to one RX slice **140**, and each has a unique carrier wavelength (color) of signal **115**. DEMUX1 **136** and DEMUX1 **137** may be for splitting each of the optical signals **129** and **119** into a number of polarized optical signals for the receiver slices **140**.

(45) Although not shown, the receiver **130** may have at least one SOA at and amplifying at least one of the signal **129**, the signal **119**, the signal **125** and/or the signal **115**. In some cases, none of these SOAs exist (as shown).

(46) A 90 degree optical hybrid **142** takes both the unmodulated signal **115** and the modulated signal **125** from the output of the PR and CR block **134** and outputs 4 optical signals **143** corresponding to the desired vectorial additions of the two input optical fields. Namely, hybrid **142** outputs two optical signals per quadrature, which may be the 4 quadrature signals I+ and I- (cosine wave quadrature signals) and Q+ and Q- (sine wave quadrature signals). In some cases, hybrid **142** outputs one optical signal per quadrature, which may be the 2 quadrature signals I+ and Q+.

(47) The hybrid **142** may be for splitting the polarization and frequency fixed optical signals **125** and **115** into 4 optical quadrature component signals (I+, I-, Q+, Q-). While the modulator **122** may only generate one optical signal **123** consisting of two quadratures I and Q; the 90-degree hybrid **142** may output 4 separate optical signals **143** by effectively splitting the (e.g., two quadrature) inputs into 4 paths and interfering the signals in those 4 paths. In the I+ path of signals **143**, no phase difference is added between the two input signals and the signals are interfered as is. In the I- path of signals **143**, a 180-degree phase difference is added between the signals and the signals are interfered. In the Q+ path of signals **143**, the phase difference is 90 degrees; and in the Q- path of signals **143** the phase difference is 270 degrees.

(48) The four output signals **143** of the hybrid **142** are directed to differential photodiodes (PDs) **144**. Each of the PDs **144** converts the optical field of one of signals **143** into an electrical photocurrent of electrical signals **145** that is proportional to the received signal power of the one signal of signals **143**. The optical field to photocurrent conversion mixes the unmodulated and modulated signals and demodulates the phase-modulated signal. This demodulation gives the QPSK phase values for the two optical signals per quadrature, I+/- and Q+/- as electrical current of signals **145**. The optical field to photocurrent conversion may be for converting at the frequencies of the data signals **117** or **121**, which may be around 10s of 10 GHz, 30 GHz for 50 Gbaud, or around 60 GHz for 100 Gbaud. This conversion may not be for conversion at the frequency of the PT signal **192** (e.g., around 1 MHz). The 4 PDs **144** may be for converting the 4 optical quadrature component signals **143** into 4 electrical current quadrature signals **145**.

(49) The photocurrent signals **145** are then converted to voltage signals **147** using TIAs **146**, which output the signals **147** to the receiver DSP **148**. TIAs **146** may be two TIAs for converting the 4 electrical current quadrature signals **145** into 4 electrical voltage quadrature data signals **147**.

(50) The receiver DSP **148** processes the voltage signals **147** received from the TIAs **146** to recover

the original data signals **117** or **121** transmitted by the TX DSP **116**. DSP **148** may be an Rx DSP for decoding or extracting the data signal from the 4 electrical voltage quadrature data signals **147**. (51) One, two or up to all four of the four hybrid output signals **143** may have an optical tap **151** that is directed to a low frequency photodiode (LF PD) of block **150** to measure the amplitude of one, two or up to four of the PR+CR marker tone signals **192** in signals **143**. Each of taps **151** may divert an inconsequential amount of the optical signals **143** with respect to the function of PDs **144**'s conversions. The taps may divert between 0.1 and 5 percent of the optical power of signals **143**.

(52) Each LF PD may convert the low frequency (e.g., around 1 MHz; or between 0.1 MHz and 10 MHz) optical field of one of signals **143** into an electrical photocurrent of electrical signal **152** that is proportional to the received signal power of the PT signal **192** of the tapped one of signals **143**. The optical field to photocurrent conversion may be for converting at the frequency of the PT signal **192**. It may not be for conversion at the frequencies of the data signals **117** or **121**. The LF PD may output across a resistor of block **150** or controller **158** to provide a voltage signal **152** that is proportional to the received signal power of the PT signal **192**. Although only one set of tap **151**, block **150** and signal **152** is shown for convenience there may be one set tapping two or each of the signals **143** and the signals **152** of those sets may be part of signal **155**.

(53) The amplitude of the PR+CR marker tone signal **192** in one, two or up to all four of signals **145** may be measured by measuring a voltage signal of one, two or up to all four dedicated transimpedance amplifiers (TIAs) **154**, one for each of the measured signals **145**. Each TIA **154** output may be monitored to measure the amplitude of the pilot tone signal **192** to perform PR+CR. Each TIA **154** may output a voltage signal **156** that is proportional to the received signal power of the PT signal **192** in one of the electrical signals **145**. Although only one TIA **154** is shown outputting signal **156** for convenience; two or up to four TIAs **154** may exist to monitor two or each of the signals **145**, and each of the TIAs output signals **156** may be part of signal **155**.

(54) Also, one, two or up to four of the TIA **146** outputs may be monitored to measure the amplitude of the pilot tone signal **192** to perform PR+CR. The amplitude of the PR+CR marker tone signal **192** in one, two or up to all four of signals **145** may be measured by measuring a voltage signal of one or more of TIAs **146**. Each TIA **146** may output a voltage signal **153** that is proportional to the received signal power of the PT signal **192** in one of the electrical signals **145**. Although only one TIA **146** is shown with its output being monitored and outputting signal **153** for convenience; two or up to all four TIAs **146** may have their outputs monitored for two or each of the signals **145**, and each of the TIAs output signals **153** may be part of signal **155**.

(55) Each of the two or up to four tone signals **192** may be at different frequencies in order to distinguish them at each of the different ones of tap **151** and blocks **150**; TIAs **154**; and/or TIAs **146**. Each of the different ones may be tuned to receive only one frequency of the signals **192** generated by tone generator **190**, and modulated onto signal **123** by modulator **122**. The signal **192** measurements by TIAs **154** and **146** may be for measuring at the frequency of the PT signal **192**. They need not be for measuring at the frequencies of the data signals **117** or **121**.

(56) Each of the three options (tap **151** and blocks **150**; TIAs **154**; and/or TIAs **146**) provide an output voltage signal with an amplitude based on the amplitude of signal **192**. These voltage signal output options are depicted as signal **152** output from blocks **150**, signal **156** output from TIA **154** and signal **153** output from one of TIAs **146**, any one or more of which are represented as feedback signal **155**. Based on the signal **155**, the amplitude of the pilot tone signal **192** is measured using a microcontroller (μ C) **158**, which may perform a PR and CR algorithm for or of block **134**. This measurement may be based on selecting any one or more of signals **152**, **156** and/or **153**. In some cases, the measurement is an aggregate or other combination of two or more of signals **152**, **156** and/or **153**. Each of the three options for measuring the amplitude of signal **192** may be a polarization detector for detecting (e.g., measuring, monitoring and/or determining) a polarization of the pilot tone signal **192** to identify a polarization of the modulated signal **129** having signal **192**,

and for outputting the pilot signal measurement signal **155** to the microcontroller **158** based on the detected polarization. Signal(s) **155** can then be used by controller **158** to determine or calculate signals **159** for controlling block **134** or **200**.

(57) Microcontroller **158** may be for sending the polarization recover signal **159** (which may also include a carrier recovery signal) to block **134** based on the received feedback signal **155**. PR and CR at block **134** is controlled or performed by microcontroller **158** through signals **159**. Controller **158** may perform a PR and CR algorithm, which the functionality of block **134** is based upon, such as using hardware, BIOIS, RAM, software or any combination thereof of controller **158**.

(58) The microcontroller **158** may output an analog signal (e.g., as signal **159**) using a digital to analog converter (DAC) or a pulse-width modulation (PWM) module. The DAC or PWM modules can be built into the microcontroller or can be external components that the microcontroller interfaces with (not shown). This analog voltage signal may either directly drive the phase shifters of block **134**, or indirectly by using external drivers that amplify the voltage and/or increase the range of the current delivered to the phase shifters. The phase shifters may be thermo-optic phase shifters or electro-optic phase shifters.

(59) FIG. **8A** is an example of equations **804** for how to obtain error signal $e(t)$ for a polarization closed-loop control block, such as used by controller **158** and/or block **134**. FIG. **8A** shows four equations **832** that can be solved to obtain error $e(t)$ for each degree of freedom (e.g., of polarization) of a transfer function for closed-loop polarization control, such as used for polarization demultiplexing and carrier recovery where the controller **158** and fiber **180** transfer functions satisfy $T_{\text{Controller}}T_{\text{Fiber}}=\pm I$. FIG. **8A** also shows components of receiver **130** with two sets of tap **151** and block **150**; and phase shifter control signals **159** from controller **158** to each of the phase shifters of block **134**.

(60) FIG. **8B** is an example of Jones matrices **806** for proportional, integral, derivative (PID) tuning of the phase shifters and couplers of block **134** using equations **804**, such as used by controller **158** and providing control of block **134** by signals **159**. Matrices **806** include phase shifter matrix **842** for tuning or controlling the phase of shifters $\Phi_{\text{sub.1}}-\Phi_{\text{sub.4}}$ and $\Phi_{\text{sub.1'}}-\Phi_{\text{sub.4'}}$. Matrices **806** also include phase coupler matrix **844** for tuning or controlling the and the phase coupling of couplers **216**, **226**, **236** and **246**.

(61) In some cases, two signals **192** are added to signal **123** by modulator **122**, one in each of quadratures I and Q of signal **123**. In these cases, the following can be measured to obtain the polarization recovery error: 1) amplitude of the two tone signals **192** in their desired channels (e.g., I tone signal **192** in I channel, and Q tone signal **192** in Q channel), 2) amplitude of the tone signals **192** in the undesired channels (I tone signal **192** in Q channel, Q tone signal **192** in I channel), and 3) The sign of the 4 tone signals **192** (in phase (0 degree) or out of phase (180 degree)). Each of these measurements can be performed (e.g., in the 2 quadratures) by one or more of tap **151** and blocks **150**; TIAs **154**; and/or TIAs **146** which output the measurements as signals **152**, **156** and/or **153** (e.g., signal **155**) to controller **158**. With this information, a linear system of 4 equations is obtained, such as by controller **158**.

(62) Solving the equations **804** or **832**, such as at controller **158**, gives the combined Jones matrix **806** of the Fiber (e.g., T_{Fiber}) and PR+CR (e.g., $T_{\text{Controller}}$). Notably, Jones matrix **842** can be used for the phase shifters and matrix **844** can be used for the (2×2) phase couplers to describe how optical elements modify the state of polarization of polarized light. Because the PR+CR phase shifter values of matrix **842** are known, another system of 4 equations can be solved to obtain the Jones matrix of only the Fiber. Each Jones matrix may contain 3 degrees of freedom, which correspond to the phase shifts set in the PR+CR block **200/134**. Therefore, three error signals can be obtained by subtracting each of the three angles $\text{Error1}=\text{FiberAngle1}-\text{PRCRAngle1}$, and so on, such as of equations **804**. The three error signals are fed into a controller, i.e. a PID controller, in order to generate the drive signal (such as signals **159**) for the phase shifters $\Phi_{\text{sub.1}}-\Phi_{\text{sub.4}}$ and $\Phi_{\text{sub.1'}}-\Phi_{\text{sub.4'}}$.

(63) Each of the two or up to four tone signals **192** may be at different frequencies in order to distinguish them at receiver **130**. By eliminating the errors of error signals $e(t)$ for each degree of freedom, both PR and CR can be performed simultaneously. So, the marker tones may be used to do PR and CR.

(64) In FIG. **1A**, both polarization recovery (PR) and carrier recovery (CR) at block **134** are performed before demultiplexing (at DEMUX1 **136** and DEMUX2 **137**). FIG. **1B** is a block diagram of a system **102** for PM-SH-ACD optical communications having polarization recovery (PR) at **164** prior to and carrier recovery (CR) at **165** after wavelength based demultiplexing at **166** and **167**. FIG. **1C** is a block diagram of a system **104** for PM-SH-ACD optical communications having polarization recovery (PR) and carrier recovery (CR) at **172** after wavelength based demultiplexing at **176** and **177**.

(65) In the three variants of a general communication link architecture shown as systems **100**, **102** and **104**, the transmitter **110** part of the link/system is identical, and the receivers **130**, **160** and **170** differ as shown. The components or blocks that are included in all of the variants were first described in a certain order for the variant of FIG. **1A**. However, the other variants FIGS. **1B** and **1C** may use the same blocks with the order of the blocks changed for each variant.

(66) In system **102**, PR is performed at block **164** before demultiplexing is performed at DEMUX1 **166** and DEMUX2 **167**. After demultiplexing, CR is performed at block **165** within each of slices **162**. In this variant, the tone signal **192** is measured in each of slices **162** and is fed back to the microcontroller **158** using feedback signals **169** from each of slices **162**. Each of signals **169** may be similar to signal **155** but be for one specific slice. Measuring the tone signal **192** in each of slices **162** and feeding back signals **155** to the microcontroller **158** using feedback signals **169** may include or be using one or more of the three options (tap **151** and blocks **150**; TIA **154**; and/or using TIAs **146**) in each slice as described for system **100**. Systems **100** and **102** are compatible with broadband optical switching. In telecommunications, broadband is the wide-bandwidth data transmission that transports multiple signals at a wide range of frequencies and Internet traffic types, which enables messages to be sent simultaneously and is used in fast internet connections. In the context of Internet access, broadband is used to mean any high-speed Internet access that is always on and faster than dial-up access over traditional analog or ISDN PSTN services.

(67) In system **104**, the demultiplexing at DEMUX1 **176** and DEMUX2 **177** is performed before both PR and CR are performed at block **174** within each of slices **172**. Here, each of slices **172** contains a PR+CR block **174**. In this variant, the tone signal **192** is measured and is fed back to the microcontrollers **178** using feedback signals **155** in each of slices **172**. Measuring the tone signal **192** in each of slices **172** and feeding back signals **155** to the microcontrollers **178** may include or be using one or more of the three options (tap **151** and blocks **150**; TIA **154**; and/or using TIAs **146**) in each slice as described for system **100**. System **104** is compatible with both broadband optical switching and wavelength-dependent optical switching. Wavelength-dependent optical switching may route (switch) signals between optical fibers on a per-wavelength basis or may use wavelength-division multiplexing (WDM) which multiplexes a number of optical carrier signals onto a single optical fiber by using different wavelengths (i.e., colors) of laser light. This technique enables bidirectional communications over a single strand of fiber, also called wavelength-division duplexing, as well as multiplication of capacity.

(68) FIG. **2** is a schematic diagram **200** of combined PR and CR block **134** that includes 8 phase shifters and 4 phase couplers. Block **134** may be a reset-free polarization controller and carrier recovery based on phase shifters.

(69) Block **134** has input signal X' (e.g., for cosine or I waves), which corresponds to one of the two received polarizations in the signal **182** at the input of block **132**, to a first phase shifter ((i) which shifts signal X' to a X'' phase shifted signal. Block **134** has input signal Y' (e.g., for sine or Q waves), which corresponds to the other of the two received polarizations in the signal **182** at the input of block **132**, to a second phase shifter ($\Phi_{\text{sub.1}}$) which shifts signal Y' to a Y'' phase shifted

signal. Then, phase coupler **216** couples the phase shifted signals from shifters (Φ .sub.1) and (Φ .sub.1') and outputs split signals **217** and **218**, where signal **217** has a 90 degree phase shift as compared to signal **218**.

(70) Next, the split signal **217** output from coupler **216** is an input signal to a third phase shifter (Φ .sub.2) which shifts signal **217** to a signal **217''** phase shifted signal and the split signal **218** is an input signal to a fourth phase shifter (Φ .sub.2') which shifts signal **218** to a signal **218''** degree phase shifted signal. Then, phase coupler **226** couples the phase shifted signals from shifters (Φ .sub.2) and (Φ .sub.2') and outputs split signals **227** and **228**, where signal **227** has a 90 degree phase shift as compared to signal **228**.

(71) Then, the split signal **227** output from coupler **226** is an input signal to a fifth phase shifter (Φ .sub.3) which shifts signal **227** to a signal **227''** degree phase shifted signal and the split signal **228** is an input signal to a sixth phase shifter (Φ .sub.3') which shifts signal **228** to a signal **228''** degree phase shifted signal. Then, phase coupler **236** couples the phase shifted signals from shifters (Φ .sub.3) and (Φ .sub.3') and outputs split signals **237** and **238**, where signal **237** has a 90 degree phase shift as compared to signal **238**.

(72) Next, the split signal **237** output from coupler **236** is an input signal to a third phase shifter (Φ .sub.4) which shifts signal **237** to a signal **237''** degree phase shifted signal and the split signal **238** is an input signal to a fourth phase shifter (Φ .sub.4') which shifts signal **238** to a signal **238''** degree phase shifted signal. Then, phase coupler **246** couples the phase shifted signals from shifters (Φ .sub.4) and (Φ .sub.4') and outputs split signal outputs of block **134** as outputs X and Y, where signal X has a 90 degree phase shift as compared to signal Y.

(73) In some cases, the PR and CR block **134** includes four sets **210**, **220**, **230** and **240**, each of 2 phase shifters (e.g., **212** and **214**) and 1 coupler (e.g., **216**), each of the 2 phase shifters for phase shifting a separate input into a separate phase shifted output, and each of the 1 coupler for coupling the separate outputs of the 2 phase shifters into a coupled signal output. In this case, separate inputs of 2 phase shifters of a first set of the four sets can be for receiving the first and second polarized optical signals; and separate inputs of 2 phase shifters of a second, third and fourth set of the four sets are for receiving the coupled signal output from the 1 coupler of a prior set of the four sets.

(74) In some cases, each pair of phase shifters (Φ .sub.X) and (Φ .sub.X') of diagram **200** introduce a relative phase between their two output paths; and the 2×2 couplers (e.g., coupler 2×6) of diagram **200** split half the power of each input arm from those two shifters equally at the output (e.g., half of the power goes to each output 2×7 and 2×8) and also introduce a 90 degree relative phase between those 2 outputs. Together, the concatenated blocks **210**, **220**, **230** and **240** allow the PR+CR block **134** to produce a reset-free arbitrary Jones matrix to recover both the polarization and recover the carrier. This Jones matrix may be Jones matrices **842** having matrix **842** for the shifters and **844** for the couplers.

(75) Diagram **200** may be divided into a separate PR section **164** and CR section **165**, such as noted for FIG. **1B**. In this case, diagram **200** can change by separating block **240** from **200** and utilizing block **240** as a standalone CR block **165** as a CR section **165** for each slice **162**, while using the sections **210**, **220**, **230** as PR block **164** for all slices **162**. Additionally, a pair of phase shifters may be used to create block **165** for CR, and another two or three phase shifters for PR. FIG. **1B**, the controller **158** may use signals **155** to calculate or create signals **159** to be sent to blocks **164** and **165**,

(76) In some cases, diagram **200** may be of combined PR and CR block **174** of FIG. **1C**. Here, diagram **200** may also be unmodified and all the sections **210**, **220**, **230**, and **240** may be used together at block **174**. For FIG. **1C**, the controller **158** may use signals **155** to calculate or create signals **159** to be sent to each of blocks **178** of slices **172**,

(77) FIG. **3** is a schematic diagram of one embodiment of an architecture **300** for PM-SH-ACD optical communications having PR and CR at block **134** prior to wavelength based demultiplexing at DEMUX1 **136** and DEMUX2 **137**. Architecture **300** may be system **100**.

(78) The electro-optic transmitter of architecture **300** is assembled on top of a substrate **320** such as a printed circuit board (PCB) or a semiconductor carrier. The entire assembly of architecture **300** is referred to as a coherent optical subassembly (COSA) **340**. The COSA consists of a receiver (RX) photonic integrated circuit (PIC) **330** and a transmitter (TX) PIC **310**, analog electronics (TIAs **146**, Drivers), DSP **148** and a μ C **158**. The COSA **340** components are interconnected using bond wires, flip-chip, vias, and/or electrical traces, such as shown by the lines connecting the different components of the architecture **300**. The PICs may be fabricated on a material such as silicon, indium phosphide (InP), or lithium niobate (LiNbO₃).

(79) This particular embodiment of architecture **300**, provides the benefits that it utilizes four multiplexed wavelengths, requires no waveguide crossings, and utilizes distributed feedback (DFB) lasers **122**. The integrated-coherent receiver (ICR) blocks **344** contain optical 90-degree hybrids **142** and PDs **144**. The IQMs **122** are IQ-modulators.

(80) Systems **102** and **104** may also be on a substrate **320**, referred to as a COSAs, have PICs, analog electronics, DSPs, μ Cs, interconnections, fabrication material, utilizes four multiplexed wavelengths, require no waveguide crossings and/or utilizes distributed feedback (DFB) lasers similar to the description for architecture **300**.

(81) FIGS. **4** and **5A-5C** are schematic diagrams of transmitter and receiver system for PM-SH-ACD optical communications that are backwards compatible with intensity modulated direct detect (IMDD) formats. FIG. **4** is a schematic diagram of a transmitter system **410** for PM-SH-ACD optical communications including laser **112** and coherent Mach-Zehnder modulators (MZM1) **401** and (MZM2) **402** that are backwards compatible with intensity modulated direct detect (IMDD) formats. IMDD may be a cost-effective communication scheme compared to coherent ones. In IMDD, the intensity, or optical power, of the light beam from a laser or a light-emitting diode (LED) is modulated by the information bits and no phase information is needed. Thus, only one quadrature I or Q is needed, not both. Also, due to this nature, no local oscillator is required for IMDD communication, which greatly eases the cost of the hardware.

(82) System **410** may be used in any of the for PM-SH-ACD optical communications systems and/or architectures herein. Modulator **422** is an IQ-modulator (IQM) used for quadrature amplitude modulation that is implemented using the two nested MZMs **401** and **402**. IQ-modulator **422** has MZM **401** for creating optical quadrature component signal I of the optical modulated data signal **123** and has MZM **402** for creating optical quadrature component signal Q of the optical modulated data signal **123**. Modulator **422** may be or be used in place of modulator **122**. Three tunable couplers **420**, **440** and **480** are added to render a conventional coherent transmitter such as TX **110** compatible with IMDD formats, such as PAM4.

(83) System **410** includes laser **112**, 6 phase shifters and 8 phase couplers, some of which are in each of tunable couplers **420**, **440** and **480**. Coupler **420** is a 2×2 coupler having phase shifter (Φ .sub.1) and phase couplers **421** and **424**. Coupler **440** is a 2×2 coupler having phase shifter (Φ .sub.2) and phase couplers **441** and **444**. Coupler **480** is a 2×2 coupler having phase shifter Φ .sub.6) and optionally also phase shifter (Φ .sub.5), and phase couplers **481** and **484**. MZM **401** may include phase shifter Φ .sub.3 and phase coupler **461**. MZM **402** may include phase shifter Φ .sub.4 and phase coupler **464**.

(84) A phase shifter Φ .sub.1 is adjusted for coupler **420** to select whether to route the light power (e.g., optical power) of laser **112** according to the desired modulation format, coherent or IMDD. For IMDD all the power is directed by coupler **420** to the modulator **422**, and for coherent modulation a fraction (e.g., half) of the power is directed to the modulator **422** and a fraction (e.g., half) of the power is directed to the LO output which may go to MUX2 **127**. Phase shifters Φ .sub.2 and Φ .sub.6 are also adjusted accordingly, with shifter Φ .sub.2 adjusted so coupler **440** selects whether to direct all the light power coupler **440** receives to one of the two nested MZMs **401** or **402** for IMDD; or direct half of the power to each of the two nested MZMs **401** and **402** for coherent modulation. Also, shifter Φ .sub.6 is adjusted so coupler **480** select whether to receive all

the light power from the directed to one of the two nested MZMs **401** or **402** for IMDD, or receive half of the power from each of the two nested MZMs **401** and **402** for coherent modulation. For IMDD only one MZM is needed because only one quadrature, I or Q, is needed to send the light power modulated communication signal, while for coherent modulation both quadratures are needed, such as for QPSK.

(85) In both cases, IMDD and coherent, the three phase shifters $\Phi_{\text{sub.1}}$, $\Phi_{\text{sub.2}}$ and $\Phi_{\text{sub.6}}$ can be set or adjusted using open or closed-loop control. In open-loop control, a lookup table can be used to set the phase of the phase shifters. In closed-loop control, a tap at the LO path and a tap before or after one of the nested MZMs are directed to monitor PDs to measure the optical power. The measured optical powers can be processed by a circuit, such as a μC (not shown but connected to each of the three phase shifters), to continuously set the correct value of phase shifting of shifters $\Phi_{\text{sub.1}}$, $\Phi_{\text{sub.2}}$, and $\Phi_{\text{sub.6}}$. For example, the phase shifters can be dithered to obtain the derivative of measured power for IMDD versus phase for coherent modulation, in order to maintain the desired power directing operating points for IMDD or coherent. Dithering is just one possible method. The phase shifters can be implemented using electro-optic phase shifters, such as thermal phase shifters, a p-type and n-type semiconductor material (PN) junction phase shifters, or any kind of material that produces an electro-optic response sufficient for this use.

(86) Consequently, for IMDD, the three phase shifters $\Phi_{\text{sub.1}}$, $\Phi_{\text{sub.2}}$ and $\Phi_{\text{sub.6}}$ can be set or adjusted so that all of laser or light power is in modulated signal **125** (e.g., signal S) and none of the power is in carrier signal **115** (e.g., signal LO). Here, all of the light power may be in only one quadrature of the modulated light power signal **125**. For coherent, the three phase shifters $\Phi_{\text{sub.1}}$, $\Phi_{\text{sub.2}}$ and $\Phi_{\text{sub.6}}$ can be set or adjusted so that the light power is in or split between modulated signal **125** (e.g., signal S) and is in carrier signal **115** (e.g., signal LO) as noted for FIGS. **1A-8C**.

(87) For coherent modulation, phase shifter (Ds of tunable coupler **480** is set to 90 degrees in order to bias the parent modulator **422** at quadrature.

(88) Optical amplifiers, such as SOAs can be utilized in the transmitter **110** to increase the optical power in order to close the optical link, i.e., to have enough power at the HF PDs **144** of the receiver **130** for those PDs to operate at the specified bit-error rate.

(89) FIGS. **5A-5C** are schematic diagrams of receiver systems **501**, **502** and **503** for PM-SH-ACD optical communications that are backwards compatible with IMDD formats. Systems **501**, **502** and **503** may be used in any of the for PM-SH-ACD optical communications systems and/or architectures herein. For example, ICRs **544**, **545** and **546** of systems **501**, **502** and **503** may include the hybrids **142** and PDs **144** of any of receivers **130**, **160** and **170** of FIGS. **1A-1C**. Also, ICRs **544**, **545** and **546** or some components thereof may be used in place of the hybrids **142** and PDs **144**, taps **151** and PDs **150** (and optionally resistors for PDs **150**) of any of receivers **130**, **160** and **170** of FIGS. **1A-1C**. One or two tunable couplers **520** and **540** are added to hybrids **142** and PDs **144** to render a conventional coherent receiver such as any or RX **130**, **160** or **170** compatible with IMDD formats, such as PAM4.

(90) The polarization controller and/or PR and CR block **134** described in FIG. **4** enables backwards compatibility from coherent modulation to IMDD formats, by setting the phase shifters ($\Phi_{\text{sub.1}}$ to $\Phi_{\text{sub.4}}$) and/or tunable couplers **420**, **440** and **480** of transmitter **410** such that all of the optical or light power is directed to the output that is connected to the S input of the ICR for IMDD. As a result of the backwards compatibility of transmitter **410**, at the hybrid **142** in FIGS. **5A-5C**, for IMDD formats, all of laser or light power is in modulated signal **125** (e.g., at input S) and none of the power is in carrier signal **115** (e.g., at input LO). Here, all of the light power may be in only one quadrature of the modulated light power signal **125/S**. For coherent modulation, the light power is in or split between modulated signal **125** (e.g., at input S) and is in carrier signal **115** (e.g., at input LO) as noted for FIGS. **1A-3**.

(91) Thus, the backwards compatibility for IMDD can be implemented at the ICRs **544**, **545** and **546** by maximizing the power detected by the LF PD that detects an optical tap of the signal

E.sub.S+E.sub.LO, such as shown in systems **501/100**, **502/102** and **503/104**, using open or closed loop control. For simplicity, only one of the four LF PDs is shown, however two or each of the four outputs of **142** may have a tap and an LF PD. For example, ICRs **544**, **545**, and **546** only have one LF PD **150**. There may be another tap **151**, LP PD **150** and signal **152** (or any of the other two measurement options) for the two tone solutions having signal **192** for each quadrature I and Q, or each of the hybrid outputs. In the particular case of IMDD mode, optical power can be measured at the IMDD path E.sub.S+jE.sub.LO (e.g., the Q quadrature can be measured by any of the three options) and the controller **158** can be used to maximize that power measurement. The polarization controller's X and Y outputs (see FIGS. **2** and **8A-C**) may be connected to the ICR's S and LO inputs. Output X may be connected to input S and output Y connected to input LO; or output X may be connected to input LO and output Y connected to input S. The IMDD data signal, similar to signal **145** but having IMDD data, can be monitored or detected at the ICRs **544**, **545** and **546** by detecting the data in the signal **155** from PD **150** from tap **151** and sending that signal to controller **158** or to RX DSP **148**. In the case of IMDD, the microcontroller can adjust the phase shifters in the PR+CR block **134** to maximize the signal power received in the LF PD, which in FIG. **5 A-C** corresponds to the E.sub.S+jE.sub.LO output of the hybrid. For IMDD, no tone signal **192** is necessary and only the total received signal power is used.

(92) ICR **544** of FIG. **5A** includes the hybrids **142**, PDs **144** and tunable coupler **520** having phase shifter $\Phi_{\text{sub.7}}$ and phase couplers **521** and **524**. Tunable coupler **520** receives signal S (e.g., signal **125**) and outputs signal **526** to input E.sub.S of hybrid **142** and/or outputs signal **528** to phase coupler **531**. Phase coupler **531** receives signal **528** and couples it with signal **143** from hybrid **142** to send as a coupled signal to input E.sub.S+E.sub.LO of PDs **144**, as shown. ICR **544** also includes tap **151** of one or multiple signals **143** to LF PD **150**, and signal **155** from PD **150** to microcontroller **158**.

(93) ICR **545** of FIG. **5B** includes the hybrids **142**, PDs **144** and tunable coupler **520**. Tunable coupler **520** receives signal S and outputs signal **526** to input E.sub.S of hybrid **142** and outputs signal **528** to new input E.sub.S of an additional HF PD **544** of PDs **144** which adds signal **528** to the I+ channel PD of PDs **144**. ICR **544** also includes tap **151** of one of signals **143** to LF PD **150**, and signal **155** from PD **150** to microcontroller **158**.

(94) ICR **546** of FIG. **5C** includes the hybrids **142**, PDs **144**, tunable coupler **520** and tunable coupler **540** having phase shifter $\Phi_{\text{sub.8}}$ and phase couplers **541** and **544**. Tunable coupler **540** receives signal **528** and outputs signal **543** to input E.sub.S+E.sub.LO of PDs **144**, as shown. ICR **544** also includes tap **151** of one of signals **143** to LF PD **150**, and signal **155** from PD **150** to microcontroller **158**.

(95) In all the variants of ICRs **544**, **545** and **546**, the optical 90-degree hybrid **142** can be bypassed by using a tunable coupler **520**, which is controlled by setting the value of phase shifter $\Phi_{\text{sub.7}}$. In coherent operation, phase shifter $\Phi_{\text{sub.7}}$ is set so that all of the optical power in the S path is routed to the E.sub.S input of the optical hybrid, as noted for FIGS. **1A-3**. In IMDD operation, phase shifter $\Phi_{\text{sub.7}}$ is set so none of the optical power in the S path is routed to the E.sub.S input of the optical hybrid **142**, but is instead routed as signal **528**.

(96) ICR **544** may use a 2×1 tunable coupler **520** to route the two paths to the I+ output of PDs **144**. Signal **526** is routed to input E.sub.S of hybrid **142** for coherent modulation or signal **528** routed directly to input E.sub.S+E.sub.LO of PDs **144** for IMDD. ICR **545** may add an additional HF PD **544** to the I+ channel PD of PDs **144**. Signal **526** is routed to input E.sub.S of hybrid **142** for coherent modulation or signal **528** routed directly to input E.sub.S of HP PD **544** and that output is added to add the HF PD output of the HF PDs **144** having input E.sub.S+E.sub.LO. ICR **546** may use coupler **520** and an additional two 2×1 tunable couplers **540**, which is controlled by setting phase shifter $\Phi_{\text{sub.7}}$ to route signal **526** to input E.sub.S of hybrid **142** for coherent modulation or to route signal **528** as signal **543** directly to input E.sub.S+E.sub.LO of PDs **144** for IMDD.

(97) Each of the tunable couplers **420**, **440**, **480**, **520** and **540** may have a first phase coupler for coupling a first and second input into a first and second output; a first phase shifter for outputting a phase shifted version of the first output; and a second phase coupler for coupling the phase shifted version and the second output.

(98) For all ICRs **544**, **545** and **546**, as described for FIG. 4 in transmitter backwards compatibility, the control of phase shifters $\Phi_{\text{sub.7}}$ and $\Phi_{\text{sub.8}}$ can be implemented via open loop control using a lookup table, or via closed-loop control by dithering phase shifter $\Phi_{\text{sub.7}}$ and $\Phi_{\text{sub.8}}$. Dithering is just one possible method.

(99) The systems, architectures and processes to achieve receiver backwards compatibility from coherent modulation to IMDD described herein is also applicable to DSP-based coherent solutions, such as “DSP-Lite” and 400ZR, in which the carrier recovery and polarization control are performed in the digital domain using DSP.

(100) Optical amplifiers, such as SOAs can be utilized in the receivers of ICRs **544**, **545** and **546** to increase the optical power in order to close the optical link, i.e., to have enough power at the HF PDs **144** and **544** to operate at the specified bit-error rate.

(101) A proposed subsequent generation 1.6 Tb/s and 3.2 Tb/s solutions may be future or backwards compatible with the 800 Gb/s and 1.6 Tb/s solutions (e.g., described herein), respectively. An architecture for 1.6 Tb/s may be based on 4-wavelength 100 Gbaud 16QAM transmission. This architecture for 1.6 Tb/s can interface with 800 Gb/s 4-wavelength 50 Gbaud 16QAM by operating both TX and RX at half the nominal data rate. An architecture for 3.2 Tb/s may be based on 8-wavelength 100 Gbaud 16 QAM transmission. This architecture for 3.2 Tb/s can interface with 1.6 Tb/s 4-wavelength 100 Gbaud 16QAM by only utilizing half of the slices of the TX and RX.

(102) FIGS. 6A-6C are block diagrams of systems **601**, **602** and **603** for PM-SH-ACD optical communications that are backwards compatible with IMDD formats. Systems **601**, **602** and **603** may be systems **100**, **102** and **104** having architecture **400** and any of architectures **501**, **502** and **503**, respectively, to be backwards compatible with IMDD formats. In some cases, systems **601**, **602** and **603** may be systems **100**, **102** and **104** having architecture **400** and architectures **503**. Each of Systems **601**, **602** and **603** may be a full transceiver block diagrams incorporating backward compatibility functions in TX and RX.

(103) Systems **601**, **602** and **603** show a PM-SH-ACD link architecture with tunable couplers **420**, **440** and **480** added to TX **110** to create TX **410** or TX **610**. System **601** shows a PM-SH-ACD link architecture with tunable couplers **520** and **540** added to RX **130** to create ICR **544** and/or RX **630**. System **602** shows a PM-SH-ACD link architecture with tunable couplers **520** and **540** added to RX **160** to create ICR **545** and/or RX **631**. System **603** shows a PM-SH-ACD link architecture with tunable couplers **520** and **540** added to RX **170** to create ICR **546** and/or RX **632**.

(104) Description of Processes

(105) FIG. 7 shows a flow chart of a process **700** flow for PM-SH-ACD optical communications. The process **700** may be performed by one or more of the systems, architectures, transmitters and/or receivers of FIGS. 1A-6C. The process **700** starts at step **710** and can end at step **790**, but the process can also be cyclical by returning to step **710** after step **790**. For example, the process may return to be performed additional times after changing data paths from or the TX DSP **116**, changing modulation at drivers **118** or changing modulation at modulator **122**.

(106) Step **710** is splitting laser light into a first and a second output of the laser light. Step **710** may include receiving a laser light having one frequency from a laser that is to be split. Step **710** may be performed by laser **112** and/or splitter **114**.

(107) Step **720** is outputting an optical modulated data signal of the first output of the laser light modulated with QPSK electrical data signals based on output channel data signals and a pilot tone signal identifying a polarization of the modulated signal. Step **720** may include receiving the electrical output channel data signals in one of various modulation formats. Step **720** may include

voltage amplifying or QPSK modulating the electrical output channel data signals into electrical QPSK data signals prior to optical modulation with the laser light. Step **720** may be performed by TX DSP **116**, drivers **118** and/or modulator **122**.

(108) Step **730** is combining into two optical signals, by rotating the polarization of one of the signals by 90 degrees and adding the optic fields of the rotated signal one of and the non-rotated one of the optical modulated data signal and the second output of the laser light. Step **730** may include polarization rotating by 90 degrees one of the optical modulated data signal or the second output of the laser light, prior to combining or adding, step **730** may include transmitting the polarized signal to a fiber optic cable. Step **730** may be performed by PBC **128**.

(109) Step **740** is polarization un-rotating by 90 degrees the polarized signal into a first polarized optical signal and a second polarized optical signal, each having at least some of the second output of the laser light and some of the optical modulated data signal. Step **740** may include that the received signal's two received polarizations are separated and then one of them is rotated by 90 degrees. Step **740** may include splitting of the two polarizations and 90-degree rotation of one of them. Step **740** may include receiving the polarized signal from the fiber optic cable. Step **740** may be performed by PSR **132**.

(110) Step **750** is recovering the original transmitted polarization of the signal **182**. Step **750** may include recovering the original transmitted polarization of the first **129** and second **119** polarized optical signals based on a received polarization recovery signal to recover a polarization fixed first optical signal, having the optical modulated data signal and to recover a polarization fixed second optical signal having the second output of the laser light, wherein the first and second polarization fixed first optical signal have the same polarization. Step **750** may be performed by block **134**, block **164** or block **174**.

(111) Steps **760** is demodulating the first and second polarizations of the optical signal **182**. Step **760** may include (a) recovering the frequency and phase of the second output of the laser light **119** from the polarization fixed second optical signal; (b) demodulating the polarization fixed first optical signal **125** into a polarization and frequency fixed first optical signal having the data signal and the pilot tone signal; and (c) demodulating the polarization fixed second optical signal into a polar, frequency and phase fixed second optical signal that is the second output of the laser light. In some case, (a) recovering the frequency and phase is not performed. Step **760** may be performed by block **134**, block **165** or block **174**.

(112) Steps **750** and **760** may be performed simultaneously by block **134** or **174**. In other cases, step **750** may be performed by block **164** and step **760** performed later by block **165**.

(113) Step **770** is splitting the polarization and frequency fixed first optical signal and the polar, frequency and phase fixed second optical signal into 4 optical quadrature component signals (I+, I-, Q+, Q-). Step **770** may also include converting the 4 optical quadrature component signals into 4 electrical current quadrature signals, and converting the 4 electrical current quadrature signals to 4 electrical voltage (e.g., QPSK) quadrature data signals. Step **770** may be performed by hybrid **142**, PDs **144** and TIAs **146**.

(114) In some cases, step **770** may include (a) recovering the frequency and phase of the second output of the laser light **119** from the polarization fixed second optical signal; (b) demodulating the polarization fixed first optical signal **125** into a polarization and frequency fixed first optical signal having the data signal and the pilot tone signal; and (c) demodulating the polarization fixed second optical signal into a polar, frequency and phase fixed second optical signal that is the second output of the laser light. In some case, (a) recovering the frequency and phase is not performed. This part of step **770** may be performed by block **134**, block **165** or block **174**.

(115) Step **780** is decoding the (e.g., QPSK) data signal from the 4 electrical voltage QPSK quadrature data signals. Step **780** may be performed by RX DSP **148**.

(116) Step **790** is detecting a polarization of the pilot tone signal(s) to identify a polarization of the modulated signal. Step **790** may include detecting a polarization of one or two tone signals **192**,

Step **790** may also include outputting the pilot signal measurement signal to the microcontroller based on the detected polarization. Step **790** may also include sending the polarization recover signal to the polarization recover component based on a received pilot signal measurement signal. Step **790** may be performed by tap **151** and blocks **150**; TIA **154**; and/or using TIAs **146** to provide an output voltage signal **155** with an amplitude based on the amplitude of signal **192**. Based on the signal **155**, the amplitude of the pilot tone signal **192** is measured using a microcontroller (μ C) **158**, which may perform a PR and CR algorithm for or of block **134**; **164** and **165**; and/or **174**.

(117) In some cases, only steps **710-730** or **740-790** are performed. In some cases, only steps **720**, **730**, **750**, **760** and **790** are performed.

CLOSING COMMENTS

(118) Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. With regard to flowcharts, additional and fewer steps may be taken, and the steps as shown may be combined or further refined to achieve the methods described herein. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

(119) As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

Claims

1. A system for polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) architecture for optical communication links, the system comprising: a receiver section having: a polarization splitter rotator (PSR) for splitting received light from an optical fiber into its two constituent polarization components; a polarization recovery (PR) component for demultiplexing the two constituent polarization components into two demultiplexed polarization signals using a pilot tone or other signal; a carrier recovery (CR) component for locking the phase of the two demultiplexed polarization signals at the output of the PR component into a first and a second phase-locked signal, the first phase-locked signal containing a modulated signal and the second phase-locked signal containing an unmodulated signal; a 90 degree optical hybrid component for producing optical quadrature components of the two phase-locked signals; photodiodes (PDs) for converting the optical quadrature components into electrical current quadrature signals; two TIAs for converting the electrical current quadrature signals to electrical voltage quadrature data signals; and a polarization detector for detecting one or more pilot tone signals used by the PR component to determine the state of polarization of the two constituent polarization components.
2. The system of claim 1, further comprising: a transmitter section having: an optical modulator for outputting the optical modulated data signal of a first output of a laser light modulated with a data signal and a pilot tone signal; and a polarization beam combiner (PBC) for: polarization rotating by

90 degrees one of the optical modulated data signal or the second output of the laser light; combining the unrotated and rotated polarization optical signals.

3. The system of claim 2, the transmitter section further comprising: a transmitter digital signal processor (TX DSP) sending output channel data signals in one of various modulation formats, the data signal based on the output channel data signals; a laser for outputting a laser light having one frequency and one phase to a splitter for creating the first and the second output of the laser light; the receiver section further comprising: a receiver digital signal processor (Rx DSP) for decoding output channel data signals from the 4 electrical voltage quadrature data signals; a microcontroller for sending a polarization recovery signal to the polarization recovery component based on a received pilot signal measurement signal; and wherein the polarization detector outputs one or more of the pilot signal measurement signals to the microcontroller.

4. The system of claim 2, the transmitter section further comprising: a plurality of transmitter slices each having: a laser with a different frequency than a laser of any other of the plurality of transmitter slices; the splitter for splitting the laser into paths for modulated and unmodulated light; drivers for creating the data signal that is different than a data signal of any other of the plurality of transmitter slices from the output channel data signals; the optical modulator for outputting the optical modulated data signal; a first multiplexer (MUX1) for combining the optical modulated data signal of the plurality of transmitter slices into a combined optical modulated data signal; a second multiplexer (MUX2) for combining the second output of the laser light of the plurality of transmitter slices into a combined optical carrier signal; wherein the PBC is for: polarizing by 90 degrees one of the optical modulated data signal or the second output of the laser light; combining the unrotated and rotated optical polarizations; the receiver section further comprising: a first demultiplexer (DEMUX1) for splitting the optical signal from the first split polarization component into a plurality of optical signals with different frequencies for a plurality of receiver slices; a second demultiplexer (DEMUX2) for splitting the optical signal from the first split polarization component into a plurality of optical signals with different frequencies for the plurality of receiver slices; the plurality of receiver slices each having: the 90 degree optical hybrid component; the 4 photodiodes (PDs); the two TIAs; and the polarization detector.

5. The system of claim 4, the transmitter section further comprising no SOAs, one SOA or a plurality of SOAs for amplifying at least one of the optical modulated data signal and the second output of the laser light, at least one of prior to or after the MUX1 and MUX2; and the receiver section further comprising no SOAs, one SOA or a plurality of SOAs for amplifying at least one constituent polarization components.

6. The system of claim 2, wherein the optical modulator is an IQ-modulator having a first Mach-Zehnder Modulator (MZM1) and a second Mach-Zehnder Modulator (MZM2) for creating optical quadrature component signals.

7. The system of claim 6, further comprising an architecture for coherent modulation that is a backwards compatible architecture for IMDD, the architecture having: a first tunable coupler between a laser and the IQ-modulator, the first tunable coupler for selecting whether to route (a) all power of the second output of the laser to the IQ-modulator for IMDD; or (b) some power of the second output of the laser to the IQ-modulator and some of the power the second output of the laser signal to the MUX2 for coherent modulation; a second tunable coupler between the first tunable coupler and the IQ-modulator for selecting whether to direct (a) all the light power the second tunable coupler receives from the first tunable coupler to one of the MZM1 or the MZM2 for IMDD; or (b) half of the power the second tunable coupler receives from the first tunable coupler to each of the MZM1 and the MZM2 for coherent modulation; and a third tunable coupler between the first and second MZMs and the MUX1 for selecting whether to (a) receive all the light power from the one of the MZM1 and the MZM2 for IMDD, or (b) receive half of the light power from each of the MZM1 or the MZM2 for coherent modulation.

8. The system of claim 7, further comprising one of: a fourth tunable coupler between DEMUX1

and the 90 degree optical hybrid component, the fourth tunable coupler for selecting whether to route all power of the optical signal to the photodiodes for IMDD or to the 90 degree optical hybrid component for coherent modulation; a fifth tunable coupler between DEMUX1 and the 90 degree optical hybrid component, the fifth tunable coupler for selecting whether to route all power of the optical signal to a fifth HF photodiode added to the 4 photodiodes for IMDD, or to the 90 degree optical hybrid component for coherent modulation; or the fourth tunable coupler and a sixth tunable coupler between the fourth tunable coupler and the 90 degree optical hybrid component and the 4 photodiodes (PDs) for selecting whether to route all power from the fourth tunable coupler to one of the photodiodes for IMDD or from the 90 degree optical hybrid component to the 4 photodiodes for coherent modulation.

9. The system of claim 8, wherein each of the first-sixth tunable couplers has a first coupler for coupling a first and second input into a first and second output; a first phase shifter for outputting a phase shifted version of the first output; zero or one second phase shifters for outputting a phase shifted version of the second output; and a second coupler for coupling the phase shifted versions of the first and second outputs.

10. The system of claim 1, wherein the PR and CR components form a combined PR and CR block comprising four sets of 2 phase shifters and 1 coupler, each of the 2 phase shifters for phase shifting a separate input into a separate phase shifted output, and each of the 1 coupler for coupling the separate outputs of the 2 phase shifters into a coupled signal output; wherein separate inputs of 2 phase shifters of a first set of the four sets are for receiving the first and second constituent polarization components of the received optical signal; and wherein separate inputs of 2 phase shifters of a second, third and fourth set of the four sets are for receiving the coupled signal output from the 1 coupler of a prior set of the four sets.

11. The system of claim 1, wherein the polarization detector includes one of: at least one optical tap having a low frequency photodiode and a resistor for reading at least one first voltage signal from at least one of the 4 optical quadrature component signals; at least one TIA for reading at least one second voltage signal of at least one of the 4 electrical voltage quadrature data signals; or at least one of the two TIAs for reading at least one third voltage signal of at least one of the 4 electrical voltage quadrature data signals; wherein the pilot signal measurement signal is one or more of the first, second or third at least one voltage signal.

12. A system for polarization-multiplexed self-homodyne analog coherent (PM-SH-ACD) architecture for optical communication links, the system comprising: a polarization splitter rotator (PSR) for splitting received light from an optical fiber into its two constituent polarization components; a polarization recovery (PR) component for demultiplexing the two constituent polarization components into two demultiplexed polarization signal using a pilot tone; a carrier recovery (CR) component for locking the phase of the two demultiplexed polarization signals at the output of the PR component into a first and a second phase-locked signal, the first phase-locked signal containing a modulated signal and the second phase-locked signal containing an unmodulated signal; a 90 degree optical hybrid component for producing optical quadrature components of the two phase-locked signals; photodiodes (PDs) for converting the optical quadrature component signals into electrical current quadrature signals; two TIAs for converting the electrical current quadrature signals to electrical voltage quadrature data signals; and a polarization detector for detecting the pilot tone signals used by the PR component to determine the state of polarization of the two constituent polarization components.

13. The system of claim 12, further comprising: an optical modulator for outputting the optical modulated data signal of a first output of a laser light modulated with a data signal and a pilot tone signal; and a polarization beam combiner (PBC) for: polarization rotating by 90 degrees one of the optical modulated data signal or the second output of the laser light.

14. The system of claim 13, further comprising: a plurality of transmitter slices each having: a laser with a different frequency than a laser of any other of the plurality of transmitter slices; the splitter

for splitting the laser into paths for modulated and unmodulated light; drivers for creating the data signal that is different than a data signal of any other of the plurality of transmitter slices from the output channel data signals; the optical modulator for outputting the optical modulated data signal; a first multiplexer (MUX1) for combining the optical modulated data signal of the plurality of transmitter slices into a combined optical modulated data signal; a second multiplexer (MUX2) for combining the second output of the laser light of the plurality of transmitter slices into a combined optical carrier signal; wherein the PBC is for: polarization rotating by 90 degrees one of the optical modulated data signal or the second output of the laser light; combining the unrotated and rotated optical polarizations; a first demultiplexer (DEMUX1) for splitting the optical signal from the first split polarization component into a plurality of optical signals with different frequencies for a plurality of receiver slices; a second demultiplexer (DEMUX2) for splitting the optical signal from the first split polarization component into a plurality of optical signals with different frequencies for the plurality of receiver slices; the plurality of receiver slices each having: the 90 degree optical hybrid component; the 4 photodiodes (PDs); the two TIAs; and the polarization detector.

15. The system of claim 13, further comprising an architecture for coherent modulation that is a backwards compatible architecture for IMDD, the architecture having: a first tunable coupler between a laser providing the laser light and the modulator, the first tunable coupler for selecting whether to route (a) all power of the laser light to the modulator for IMDD; or (b) some power of the laser light to the modulator and some of the power the laser light to the MUX2 for coherent modulation; a second tunable coupler between the first tunable coupler and the modulator for selecting whether to direct (a) all the light power the second tunable receives from the first tunable coupler to one of the MZM1 or the MZM2 for IMDD; or (b) half of the power the second tunable coupler receives from the first tunable couple to each of the MZM1 and the MZM2 for coherent modulation; and a third tunable coupler between the first and second MZMs and the MUX1 for selecting whether to (a) receive all the light power from the one of the MZM1 or the MZM2 for IMDD, or (b) receive half of the light power from each of the MZM1 and the MZM2 for coherent modulation.

16. The system of claim 15, further comprising one of: a fourth tunable coupler between DEMUX1 and the 90 degree optical hybrid component, the fourth tunable coupler for selecting whether to route all power of the optical signal to the photodiodes for IMDD or to the 90 degree optical hybrid component for coherent modulation; a fifth tunable coupler between DEMUX1 and the 90 degree optical hybrid component, the fifth tunable coupler for selecting whether to route all power of the optical signal to a fifth HF photodiode added to the 4 photodiodes for IMDD, or to the 90 degree optical hybrid component for coherent modulation; or the fourth tunable coupler and a sixth tunable coupler between the fourth tunable coupler and the 90 degree optical hybrid component and the 4 photodiodes (PDs) for selecting whether to route all power from the fourth tunable coupler to one of the photodiodes for IMDD or from the 90 degree optical hybrid component to the 4 photodiodes for coherent modulation.
